Many Alerts, Few Injuries:

Alert Burden and Yield of Common Training-Load and Wellness Rules in Elite Women's Football

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Preprint. This manuscript has not been peer reviewed. It is intended for submission to a peer-reviewed journal.

Original article | Word count: 3,200 (main text) | 3 tables, 4 figures

Abstract

Athlete monitoring systems often turn training-load and wellness data into simple alerts, but little is known about how many alerts these rules produce or how often an alert is followed by injury. This study applied four common rules to two seasons of self-reported data from two elite Norwegian women's football teams (SoccerMon dataset; 50 players, 2020–2021): an acute:chronic workload ratio (ACWR) above 1.5, a week-to-week load increase above 15%, a wellness z-score of -1.5 or lower, and training monotony above 2.0. Alert burden was expressed as new alerts per week for a 25-player squad, and alert yield as the percentage of new alerts followed by a self-reported injury within 7 days (29 monitored injury onsets in 11 players). The weekly-load rule fired on 35% of eligible days and produced a median of 16.4 new alerts per squad-week; the other rules produced about 3.5 each. Only 0.2–1.2% of new alerts were followed by an injury report, close to the 0.9% baseline on any monitored day, and confidence intervals were wide. Simple rules therefore created a substantial workload for staff, while no rule's yield was clearly above the baseline in this sample.

Keywords: *soccer, athlete monitoring, acute:chronic workload ratio, wellness, injury, decision support*

1 Introduction

Most professional football clubs now monitor their players routinely, collecting training load, often as session rating of perceived exertion (sRPE), and short wellness questionnaires (Akenhead and Nassis, 2016; Saw et al., 2016). To make these data usable day to day, they are often condensed into simple rules that flag a player when a value crosses a threshold, often shown as a red, amber or green status on a dashboard (Robertson et al., 2017; Thornton et al., 2019). Typical rules include an acute:chronic workload ratio (ACWR) above about 1.5 (Hulin et al., 2016; Gabbett, 2016), a week-to-week increase in load of more than 10–15% (Gabbett, 2016), a wellness score well below the player's own usual values, often expressed as a z-score (Thornton et al., 2019), and high training monotony (Foster, 1998).

Each of these rules has been studied mainly as a risk factor: is the injury rate higher when the rule is true? This is a different question from the one faced by practitioners, who must decide what to do each time a flag appears. For that decision, two other properties matter. The first is *alert burden*: how many alerts a rule produces for a squad in a typical week. The second is *alert yield*: of the alerts raised, how many are followed by the event they are meant to warn about. A rule can be associated with injury at group level and still be of little practical use if it fires so often that most alerts are false alarms (Bahr, 2016; Fanchini et al., 2018). In clinical settings, frequent non-actionable alarms are known to lead to "alarm fatigue", in which staff become desensitised and real warnings are missed (Sendelbach and Funk, 2013). Practitioners in football already report that monitoring is less effective in preventing injury than they expect it to be (Akenhead and Nassis, 2016).

The ACWR in particular has been criticised on conceptual and statistical grounds, including mathematical coupling between its numerator and denominator and the arbitrary nature of its thresholds (Lolli et al., 2019; Impellizzeri et al., 2020b,a). Much of this debate concerns causal inference. Far less attention has been paid to the simpler operational question of how many alerts these rules generate, and how often those alerts are followed by injury, when they are applied exactly as a practitioner would apply them. In addition, much of the evidence comes from men's sport, and few openly available datasets exist for women's football.

The aims of this study were therefore to (i) quantify the alert burden of four common training-load and wellness rules in elite women's football, and (ii) estimate the alert yield of each rule, that is, the proportion of new alerts followed by a self-reported injury within 7 days, compared with the baseline injury rate on any monitored day. Because the number of injuries in any single squad is small, the second aim was treated as exploratory. Burden and yield are simple, practitioner-facing measures that can be reported for any monitoring rule, and a secondary purpose of this study was to show how they can be calculated from routinely collected data.

2 Methods

2.1 Data

This study used the subjective data of the SoccerMon dataset (Midoglu et al., 2024), which were collected with the PmSys athlete monitoring system from two teams in the Norwegian women's elite league (Toppserien) during the 2020 and 2021 seasons, and are released under a CC-BY 4.0 licence. Fifty players reported subjective data (Team A, 27; Team B, 23). After each session, players reported its duration

and perceived exertion (RPE, 1–10). Once a day, they reported fatigue, mood, sleep quality, muscle soreness and stress (each 1–5) and readiness to train (1–10), with higher values indicating a better state. Players could also report injuries at any time by marking one or more of 22 body locations on a body map as minor or major; these reports are self-reported and do not include a medical diagnosis (Midoglu et al., 2024). The data were arranged as one row per player per calendar day (731 days, 1 January 2020 to 31 December 2021; 36,550 player-days).

2.2 Training load and wellness variables

Daily load was the dataset's daily training-load value, which is derived from session RPE and duration (session RPE, sRPE; arbitrary units) (Foster et al., 2001; Midoglu et al., 2024); days without a load entry were given a load of zero. All load variables were computed over windows that ended on, and included, the current day. Weekly (acute) load was the sum of daily load over 7 days. Chronic load was the mean daily load over 28 days, and the ACWR was the mean daily load over 7 days divided by chronic load. Because the 7-day window lies inside the 28-day window, this is the conventional "coupled" ACWR (Lolli et al., 2019), which was used deliberately because it is the version most often applied in practice. Training monotony was the mean daily load over 7 days divided by its standard deviation (population form) (Foster, 1998). For wellness, each of the six items was converted to a z-score against the player's own mean and standard deviation over the previous 28 days (requiring at least 10 entries, and excluding the current day), and the six z-scores were averaged to give a composite score.

2.3 Alert rules

Four rules were chosen because they are widely used in applied practice (Table 1); the ACWR and weekly-load thresholds come from the literature, whereas the wellness and monotony thresholds reflect common practice rather than a single validated cut-off. For each rule, a player-day was *eligible* only when the inputs needed to compute the rule were available, so that missing data were not counted as normal days. An *alert day* was an eligible day on which the rule was true. Because a rule often stays true for several consecutive days, the analysis counted *new alerts*: the first day of each run of consecutive alert days. This mirrors practice, where a new flag prompts action and a flag that persists usually does not prompt a fresh one. A fifth rule, "any rule", raised an alert when at least one of the four rules was true.

2.4 Injury outcome

Because SoccerMon does not merge reports from consecutive days (Midoglu et al., 2024), repeated reports were grouped into episodes: a new episode started when more than 7 days had passed since the player's previous report. The first day of each episode was taken as the injury onset. This gave 40 onsets in 15 players (36 in Team A; 11 including at least one major report). To make sure that an onset could in principle have been preceded by an alert, the primary analysis included only *monitored onsets*, defined as onsets with at least 3 eligible days in the preceding 7 days. Eleven onsets were excluded in this way, including seven reported on 20–22 March 2020, during the early weeks of the COVID-19 lockdown, when little or no training had been recorded in the preceding week. The primary outcome sample was therefore 29 monitored onsets in 11 players (27 in Team A). For each player-day, the outcome was

Table 1: Alert rules and eligibility criteria.

Rule	Alert when	Basis
ACWR > 1.5	7-day mean load divided by 28-day mean load exceeds 1.5; eligible from 28 days after the player's first load entry, when chronic load and 7-day load are both > 0	Hulin et al. (2016); Gabbett (2016)
Weekly load +15%	7-day load is more than 15% higher than the 7-day load one week earlier; eligible when both weeks have load > 0	Gabbett (2016)
Wellness z ≤ -1.5	Composite wellness z-score is -1.5 or lower; eligible when a valid baseline (at least 10 entries in the previous 28 days) exists	Individual z-score approach (Thornton et al., 2019); threshold set for this study
Monotony > 2.0	7-day mean load divided by its standard deviation exceeds 2.0; eligible when 7-day load > 0	Monotony as defined by Foster (1998); threshold common in practice
Any rule	At least one of the four rules is true	–

positive if a monitored onset occurred in the following 7 days (days $t+1$ to $t+7$).

2.5 Outcome measures

Alert burden was expressed in two ways: the percentage of eligible player-days that were alert days, and the number of new alerts per week, scaled to a squad of 25 players (number of new alerts in a week divided by the number of players with at least one eligible day that week, multiplied by 25; weeks with fewer than 5 eligible players were excluded). The percentage of player-weeks with at least one new alert was also reported. *Alert yield* was the positive predictive value (PPV): the percentage of new alerts followed by a monitored onset within 7 days. The *baseline* was the same percentage for all eligible player-days of that rule, which is the yield expected if alerts were raised at random. The risk ratio (RR) compared the outcome rate on alert days with that on eligible non-alert days.

2.6 Ethics

This was a secondary analysis of publicly released, anonymised data, made available under a CC-BY 4.0 licence (Midoglu et al., 2024). No new data were collected and no participants were contacted; ethical approval was therefore not required. The analysis was conducted in line with the principles of the Declaration of Helsinki.

2.7 Sample size estimation and justification

The sample was fixed by the available data, so an a priori sample size calculation was not possible. Following Ditroilo et al. (2025), a sensitivity analysis was conducted instead. With a one-sided exact binomial test at $\alpha = 0.025$ and 80% power, and treating new alerts as independent, the smallest PPV that could be distinguished from the baseline was about 2.5% for ACWR (450 new alerts; RR about 2.9), 1.6% for the weekly-load rule (2,033; RR 1.8), 2.6% for the wellness rule (436; RR 3.1), 2.5% for monotony (500; RR 2.7) and 1.6% for any rule (2,256; RR 1.7). Because alerts from the same player are not independent, these values are optimistic. The study could therefore detect only a strong injury signal, and the yield analysis was planned as exploratory. The burden analysis, which does not depend on the number of injuries, was the primary analysis.

2.8 Statistical analysis

Ninety-five per cent confidence intervals (CI) for PPV and RR were obtained with 2,000 bootstrap resamples of players (cluster bootstrap, percentile method), so that the repeated observations within each player were respected. To test whether alerts were followed by injury more often than expected by chance, a permutation test was used: within each player, the injury onsets were moved to randomly chosen days on which that player was monitored, the PPV of each rule was recalculated, and this was repeated 2,000 times. The one-sided p value was the proportion of permuted PPVs at least as high as the observed PPV (with the usual $+1$ correction). Five sensitivity analyses were carried out: excluding the two players with the most injury episodes, using only the first onset per player, using 14-day and 28-day gaps to define episodes, and analysing Team A only. When players were excluded, all of their data were removed, so that PPV and baseline were always calculated on the same players. As a reproducibility check, all load variables were recomputed independently from the daily loads and compared with the values used in the analysis. In addition, the threshold of each rule was varied (ACWR 1.3, 1.5 and 2.0; weekly increase 10, 15 and 30%; wellness -1.0, -1.5 and -2.0; monotony 1.5, 2.0 and 2.5), with Wilson 95% CIs for PPV. Analyses were conducted in Python (pandas, NumPy, SciPy). Use of a generative AI tool for coding and writing support is described in the Declarations.

3 Results

3.1 Data available

Across both seasons, 21,332 player-days (58.4% of all player-days) were eligible for at least one rule, covering 3,209 player-weeks. The number of eligible days ranged from 15,695 for the wellness rule to 21,121 for monotony (Table 2). Injury reports were concentrated in a few players: two players accounted for 16 of the 40 episodes, and 12 of the 29 monitored onsets.

3.2 Alert burden

The rules differed greatly in how often they fired (Table 2, Figure 2). The weekly-load rule was true on 35.4% of eligible days and produced 2,033 new alerts, a median of 16.4 new alerts per week for a 25-player squad; 56.1% of player-weeks contained at least one new alert from this rule. ACWR, wellness and monotony each produced a median of about 3.5 new alerts per squad-week, and 13–15% of player-weeks contained at least one new alert from each of them. Combining all four rules gave a median of 16.7 new alerts per squad-week, with at least one new alert in 57.0% of player-weeks. Figure 1 shows how these alerts appear for one player over one season: from mid-August to the end of November, long runs of ACWR and weekly-load alerts occurred without any injury report, while the wellness and monotony rules fired only once each in the whole period.

3.3 Alert yield

Few alerts were followed by an injury report (Table 2, Figure 3). The PPV ranged from 0.22% for ACWR (1 of 450 new alerts) to 1.15% for the wellness rule (5 of 436). The baseline on any monitored day was 0.85–0.93%, so in absolute terms an alert raised the chance of an injury report in the following week by at most about 0.3 percentage points. For every rule except ACWR, the CI for PPV included the

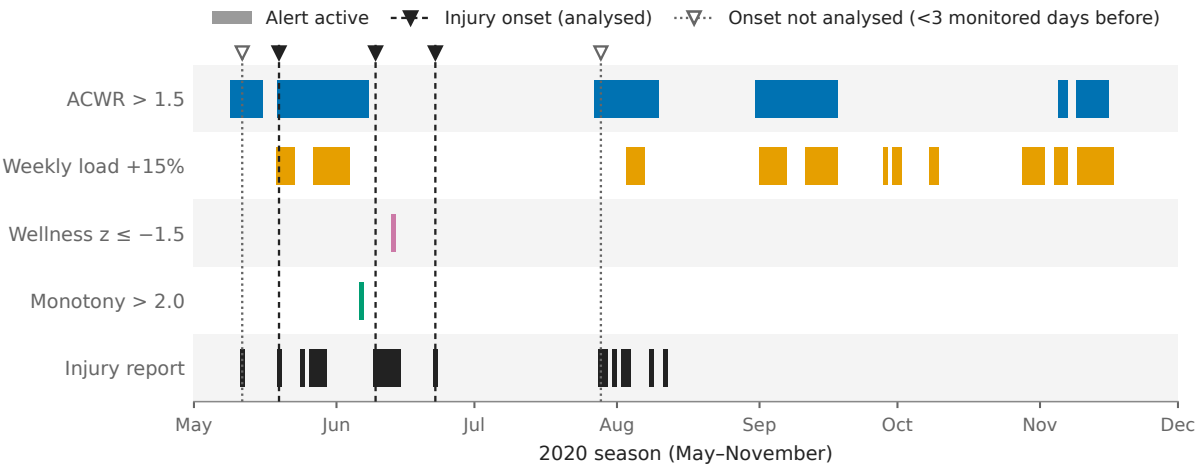


Figure 1: Alerts and injury reports for one player (one of the two players with the most monitored onsets), May to November 2020. Coloured bars show days on which each rule was true, and the bottom row shows days with an injury report. Dashed lines with filled markers show injury onsets included in the analysis; dotted lines with open markers show onsets that were excluded because the player had fewer than 3 monitored days in the preceding week.

Table 2: Alert burden and alert yield of each rule (primary analysis: 29 monitored onsets).

	Burden			Yield			
Rule	**Eligible days**	**Alert days (%)**	**New alerts per squad-week[a]**	**Followed / new alerts**	**PPV, % (95% CI)**	**Baseline (%)**	**RR (95% CI)**
ACWR > 1.5	19,895	11.5	3.5	1 / 450	0.22 (0.0–0.7)	0.87	0.83 (0.00–2.08)
Weekly load +15%	19,750	35.4	16.4	18 / 2,033	0.89 (0.3–1.6)	0.93	0.93 (0.60–1.44)
Wellness z ≤ −1.5	15,695	3.0	3.5	5 / 436	1.15 (0.2–2.4)	0.85	1.25 (0.22–2.63)
Monotony > 2.0	21,121	7.5	3.5	5 / 500	1.00 (0.2–2.1)	0.92	1.65 (0.43–3.39)
Any rule	21,332	42.0	16.7	16 / 2,256	0.71 (0.3–1.2)	0.92	1.09 (0.68–1.65)

PPV, positive predictive value (new alerts followed by a monitored injury onset within 7 days); baseline, the same percentage across all eligible days; RR, risk ratio of the 7-day outcome on alert days versus eligible non-alert days. CIs from 2,000 player-cluster bootstrap resamples.

[a] Median number of new alerts per week, scaled to a 25-player squad.

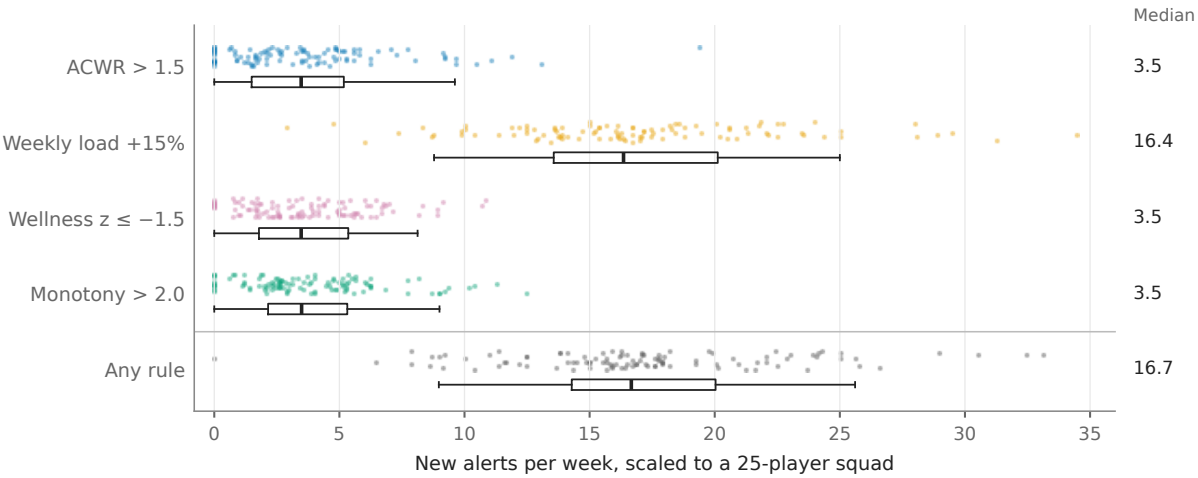


Figure 2: Alert burden. Each point is one week; values are new alerts scaled to a 25-player squad. Boxes show the median and interquartile range, and whiskers the 5th to 95th percentiles. The median for each rule is given on the right.

baseline; for ACWR, the whole interval lay below it. Every CI for RR included 1. In the permutation test, no rule was followed by injury more often than expected by chance (one-sided $p = 0.16$ for wellness, 0.47 for weekly load, 0.83 for monotony, 0.91 for any rule and 0.99 for ACWR). Put another way, staff would have needed to act on about 87 wellness alerts, 100 monotony alerts or 113 weekly-load alerts to meet one player who went on to report an injury within a week.

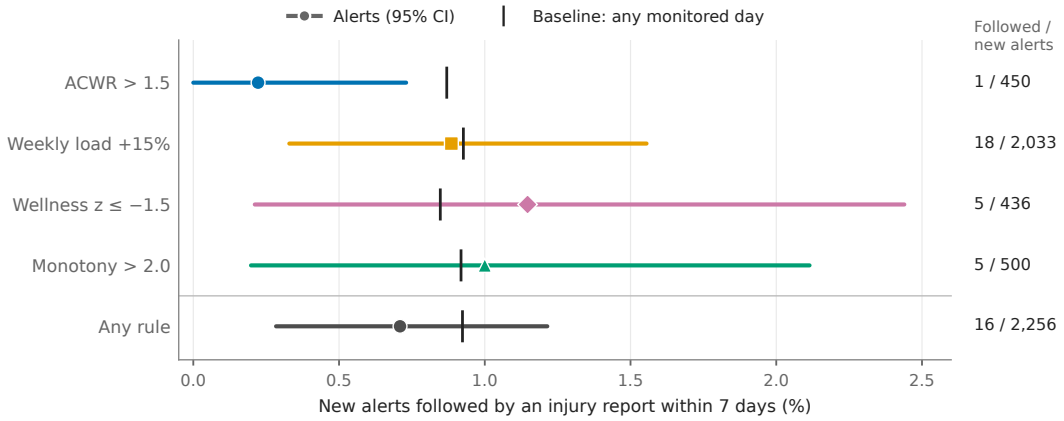


Figure 3: Alert yield. Points show the percentage of new alerts followed by a self-reported injury onset within 7 days, with 95% confidence intervals from a player-cluster bootstrap. The black tick shows the baseline for any eligible day. Numbers on the right give alerts followed by injury and the total number of new alerts.

3.4 Sensitivity analyses

The results did not change in any meaningful way across the five sensitivity analyses (Table 3). In every scenario, the PPV of each rule stayed within 0.8 percentage points of that scenario's baseline; the largest difference was for the wellness rule in Team A only (2.02% against a baseline of 1.20%). Point estimates of RR varied from 0.36 to 2.16, but these estimates were based on 11–29 onsets and were therefore unstable. Changing thresholds altered burden much more than yield (Figure 4). Stricter thresholds reduced the number of alerts, as expected, but yield did not rise in a consistent way: no alert at ACWR > 2.0 (0 of 188) or wellness ≤ -2.0 (0 of 202) was followed by injury. The only exception was monotony > 2.5, where 6 of 197 new alerts (3.0%; Wilson 95% CI 1.4–6.5%) were followed by an injury report; given the number of comparisons and the small count, this is best regarded as a possible chance finding that needs testing in other data.

The reproducibility check reproduced the stored load variables for 47 of the 50 players (99.97% of their player-days). For three players (all in Team A), the stored variables did not match the daily loads on 850 of their 2,193 player-days, which suggests a duplicated or misaligned record in the source data for these players. Excluding them (leaving 20 monitored onsets in 9 players) did not change the findings: alert days made up 3.0–41.8% of eligible days (3.0–42.0% in the full sample), and PPV ranged from 0.23% to 0.78% against baselines of 0.51–0.69%.

4 Discussion

This study examined four common monitoring rules from the point of view of the practitioner who receives the alerts. There were two main findings. First, the rules produced a large and uneven alert

Table 3: Sensitivity analyses: alert yield (PPV, %) and risk ratio by scenario.

Scenario	Onsets	Baseline, %	ACWR	Weekly load	Wellness	Monotony	Any rule
Primary analysis	29	0.85–0.93	0.22 (0.83)	0.89 (0.93)	1.15 (1.25)	1.00 (1.65)	0.71 (1.09)
Excluding top two reporters	17	0.50–0.60	0.00 (0.43)	0.62 (1.04)	1.18 (2.16)	0.67 (1.57)	0.52 (1.13)
First onset per player	11	0.25–0.35	0.00 (1.68)	0.30 (1.02)	0.46 (1.33)	0.00 (0.36)	0.27 (1.00)
Episode gap 14 days	18	0.52–0.60	0.22 (0.47)	0.74 (1.46)	0.69 (1.09)	0.80 (2.07)	0.53 (1.47)
Episode gap 28 days	16	0.45–0.53	0.22 (0.55)	0.64 (1.66)	0.69 (1.29)	0.60 (1.92)	0.49 (1.60)
Team A only	27	1.20–1.36	0.40 (1.01)	1.33 (0.93)	2.02 (1.53)	1.39 (1.45)	1.07 (1.13)

Values are PPV (%) with the risk ratio in brackets. Baseline is the range across the five rules. In each scenario, excluded players' data were removed entirely.



Figure 4: Effect of the alert threshold on yield. Points show the percentage of new alerts followed by a self-reported injury onset within 7 days, with Wilson 95% confidence intervals; the dashed line shows the baseline. Labels on the x-axis give the threshold and, below it, the number of alerts followed by injury and the number of new alerts.

burden: the weekly-load rule alone flagged more than half of all player-weeks and generated about 16 new alerts per week for a typical squad. Second, alerts were rarely followed by an injury report, with yields of 0.2–1.2%, none clearly above the baseline of about 0.9%. Burden was clear and stable; the injury signal, in this sample, was not.

4.1 Why burden matters

Alert burden is seldom reported, yet it determines whether a rule can be used. A medical or performance team that receives 16 new flags a week from one rule must either investigate each one, which takes time, or learn to ignore most of them, which defeats the purpose of the system. The clinical literature on alarm fatigue shows that the second response is common when most alarms are not actionable (Sendelbach and Funk, 2013). The weekly-load rule probably fired so often because weekly sRPE in football varies naturally with the match schedule and the training cycle, so a 15% change is a normal week-to-week fluctuation rather than an exceptional event. A rule that is true on more than one-third of eligible days cannot pick out the few days that matter.

4.2 Why yield was low

Two explanations are likely. The first is arithmetic. When the outcome is rare, even a rule with a genuine association will have a low PPV, because most alerts fall on players who would not have been injured in any case (Bahr, 2016). With a baseline of 0.9% per week, a rule would need to nearly triple the risk to reach a PPV of 2.5%, which would still mean 39 false alarms for every alert followed by injury. Previous studies in men's football have reached a similar conclusion: the ACWR was associated with injury but predicted it poorly (Fanchini et al., 2018). The second explanation concerns the rules themselves. The ACWR and similar ratios have well-described statistical weaknesses (Lolli et al., 2019; Impellizzeri et al., 2020b), and the relationship between load and injury is likely to be non-linear and to depend on many other factors (Windt and Gabbett, 2017; Andersen et al., 2021). A single threshold applied to a single variable cannot capture this. The absence of a clear signal here does not show that load or wellness are unrelated to injury; this study was not designed or powered to test that. It does show that, used as stand-alone alerts, these rules gave staff little useful warning in this setting.

4.3 Practical recommendations

Four recommendations follow for practitioners who use rule-based alerts. (1) Report and review the alert burden of every rule before using it, in the same way as its threshold. (2) Avoid rules that are true on a large share of days, such as a 15% weekly change in sRPE, unless they are combined with other information. (3) Treat an alert as a prompt for a conversation with the player rather than as a prediction of injury (Robertson et al., 2017; Thornton et al., 2019). (4) Where possible, audit the yield of the rules in the team's own data at the end of each season, so that rules that rarely lead to useful action can be retired.

4.4 Strengths and limitations

The main strengths of this study are the use of an openly available dataset from elite women's football, the practitioner-focused measures of burden and yield, several sensitivity analyses, and open code that

allows every result to be reproduced. The study also has important limitations. The injury outcome was a self-reported injury (pain) report, not a medically diagnosed time-loss injury as defined in consensus statements (Fuller et al., 2006); some reports probably describe minor complaints, and some injuries may not have been reported. The number of monitored onsets was small (29, in 11 players, almost all in Team A), so the yield estimates are imprecise and the study could detect only a strong signal. Training load was based on sRPE only, because GPS data were not used, and match load may be incomplete. Days without a load entry were treated as rest days with zero load, which is common in practice but may misclassify days on which a player trained without reporting; for the same reason, the eligibility rules may favour more compliant players. For three players, the stored load variables could not be fully reproduced from the daily loads, although excluding them did not change the results. Finally, the thresholds were taken from the literature rather than tuned to these players; individual thresholds might perform differently.

4.5 Future work

Larger, multi-team datasets with medically recorded injuries are needed to estimate alert yield with useful precision. Future studies could also test whether combining load, wellness and contextual information improves yield without increasing burden, and whether reporting burden and yield alongside conventional risk estimates changes how practitioners use monitoring data.

5 Conclusion

In two seasons of elite women's football data, four common monitoring rules generated many alerts, with the weekly-load rule alone producing about 16 new alerts per squad each week. Fewer than 1.2% of alerts were followed by a self-reported injury within a week, close to the baseline rate. Practitioners should consider the alert burden of a rule as carefully as its threshold, and should not interpret single-threshold alerts as injury predictions.

Acknowledgements

The author thanks Midoglu et al. (2024) and the players and staff of the two teams for making the SoccerMon dataset openly available.

Disclosure of interest

The author is founder and director of Ball2Data Ltd, a sports technology company working in performance analysis and athlete monitoring, and is preparing patent applications in athlete-monitoring technology. Ball2Data Ltd had no role in the design, analysis or interpretation of this study.

Funding

No funding was obtained for this work.

Data availability statement

The SoccerMon dataset is openly available on Zenodo (record 10033832) under a CC-BY 4.0 licence (Midoglu et al., 2024). All analysis code and the derived player-day data are available on GitHub (https://github.com/yashwanthajjilapuram123/paper2-alert-burden) under an MIT licence.

Declaration of generative AI use

Claude (Anthropic; Claude Opus 5.5, used in October 2026) was used to assist with writing and debugging the analysis and figure code, and with drafting and editing the manuscript text, in order to speed up programming and improve clarity. The author directed the study and made all design decisions, ran all analyses, checked every reported result against the source outputs and every reference against its original source, and takes full responsibility for the content.

Author contributions (CRediT)

Y.A.: conceptualisation, methodology, software, formal analysis, visualisation, writing – original draft, writing – review and editing.

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